

Two ledger reductions: Erdős #727 and #374

Samuel Lavery

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Abstract

We record elementary but apparently unrecorded reductions of two Erdős problems to statements about a single ledger. For #727 — does $(n+k)!^2 \mid (2n)!$ infinitely often? — we show the divisibility is exactly a carry inequality on each prime, and deduce unconditionally that any solution forces $(n+1) \cdots (n+k)$ to be $\sqrt{2n}$ -smooth; this is a genuine constraint, not a heuristic, and it is what makes the problem a smooth-number question. For #374 we show that $F(m)$ fails to exist exactly because a prime introduces a fresh coordinate into an $\mathbb{F}_2$ -valued ledger, which re-derives the known fact that no D_k contains a prime in one line. All statements here are **proved**; the accompanying densities are recorded separately as measurements.

1 Notation

For a prime p , v_p is the p -adic valuation, $s_p(m)$ the base- p digit sum, and

$$\kappa_p(n) := v_p \binom{2n}{n} = \frac{2s_p(n) - s_p(2n)}{p-1}$$

the number of carries when n is added to itself in base p (Kummer; Legendre for the middle equality).

2 Erdős #727

Theorem 1 (exact restatement). *For all $n \geq k \geq 1$,*

$$(n+k)!^2 \mid (2n)! \iff [(n+1)(n+2) \cdots (n+k)]^2 \mid \binom{2n}{n} \iff \kappa_p(n) \geq 2v_p((n+1) \cdots (n+k)) \quad \forall p.$$

Proof. $\frac{(2n)!}{(n+k)!^2} = \frac{(2n)!}{n!^2} \cdot \frac{n!^2}{(n+k)!^2} = \binom{2n}{n} / [(n+1) \cdots (n+k)]^2$, which gives the first equivalence; the second is comparison of p -adic valuations, using $v_p \binom{2n}{n} = \kappa_p(n)$. $\square$

Remark 2. *At $k = 1$ this reads $(n+1)^2 \mid \binom{2n}{n}$, which is Balakran's theorem [1]; so the reformulation is anchored against a known result. Note also that only primes dividing $(n+1) \cdots (n+k)$ impose any condition, so the criterion is a finite check.*

Lemma 3 (carry ceiling above the square root). *If $p > \sqrt{2n}$ then $\kappa_p(n) \leq 1$.*

Proof. From $p > \sqrt{2n}$ we get $p^2 > 2n > n$, so $n < p^2$ and n has at most two base- p digits; carries in $n + n$ can occur only out of positions 0 and 1. A carry out of position 1 would require $2n \geq p^2$, contradicting $p^2 > 2n$. Hence at most one carry. $\square$

Theorem 4 (smoothness is forced). *Let $k \geq 2$. If $(n+k)!^2 \mid (2n)!$ then every prime factor of $(n+1)(n+2) \cdots (n+k)$ is at most $\sqrt{2n}$; that is, the product is $\sqrt{2n}$ -smooth.*

Proof. Suppose $p \mid (n+1) \cdots (n+k)$ with $p > \sqrt{2n}$. Then $v_p((n+1) \cdots (n+k)) \geq 1$, so Theorem 1 demands $\kappa_p(n) \geq 2$, contradicting Lemma 3. $\square$

Measured (census to $n = 1.2 \times 10^5$): among all solutions for $k = 2$ and $k = 3$ there is **no** violation of Theorem 4, and $\max \log p_{\max} / \log(2n) = 0.4999$ — the bound is attained and never crossed. Solution densities to $n = 2 \times 10^5$ are 0.1126 ($k=1$), 0.0099 ($k=2$), 7.6×10^{-4} ($k=3$), 4×10^{-5} ($k=4$), with dyadic counts for $k = 2$ stabilising near 0.010: positive density, not merely infinitude. The density of n for which $(n+1)(n+2)$ is $\sqrt{2n}$ -smooth measures 0.0932, against $\rho(2)^2 = (1 - \log 2)^2 = 0.0942$ — consecutive smoothness is independent to 1%.

3 A lower bound on n in terms of k

Theorem 4 bounds the size of the rails; the carry ceiling bounds their multiplicities, and the two together bound k .

Lemma 5 (carry ceiling). *For every prime p and every $n \geq 1$, $\kappa_p(n) \leq \lfloor \log_p(2n) \rfloor$.*

Proof. $\kappa_p(n)$ counts the carries in the base- p addition $n + n = 2n$. Carries occur only out of positions $0, \dots, \lfloor \log_p(2n) \rfloor - 1$, since a carry out of the top position of $2n$ is impossible. $\square$

Lemma 6 (multiplicity ceiling). *Suppose $(n+k)!^2 \mid (2n)!$ and let p be a prime with $e := v_p((n+1) \cdots (n+k)) \geq 1$. Then*

$$e \leq \frac{1}{2} \log_p(2n), \quad \text{equivalently} \quad p^e \leq (2n)^{1/2}.$$

Proof. By Theorem 1 the hypothesis gives $\kappa_p(n) \geq 2e$, and Lemma 5 gives $\kappa_p(n) \leq \log_p(2n)$. Combining, $2e \leq \log_p(2n)$; exponentiating base p gives $p^e \leq (2n)^{1/2}$. $\square$

Verified: all 1515 rails occurring in the 227 solutions with $k \in \{2, 3\}$ and $n < 30000$ — no violation.

Theorem 7 (k is small compared with $\sqrt{n}$). *If $(n+k)!^2 \mid (2n)!$ with $k \geq 2$, then*

$$n^k < (2n)^{\pi(\sqrt{2n})/2}, \quad \text{hence} \quad k < (1 + o(1)) \frac{\sqrt{2n}}{\log n} \quad \text{and} \quad n \gg k^2 \log^2 k.$$

Proof. Write $\Pi = (n+1)(n+2) \cdots (n+k)$. By Theorem 4 every prime factor of Π is at most $\sqrt{2n}$, and by Lemma 6 each contributes $p^{v_p(\Pi)} \leq (2n)^{1/2}$. Hence

$$\Pi = \prod_{p \leq \sqrt{2n}} p^{v_p(\Pi)} \leq (2n)^{\pi(\sqrt{2n})/2}.$$

On the other hand $\Pi > n^k$, which gives the displayed inequality. Taking logarithms, $k \log n < \frac{1}{2} \pi(\sqrt{2n}) \log(2n)$, and the prime number theorem $\pi(x) \sim x/\log x$ yields $\pi(\sqrt{2n}) \sim 2\sqrt{2n}/\log(2n)$, so $k \log n < (1 + o(1))\sqrt{2n}$. Solving for n gives $n \gg k^2 \log^2 k$. $\square$

Verified: the inequality $k < \sqrt{2n}/\log n$ holds on all 302 solutions with $2 \leq k \leq 5$, $n < 40000$, as does the intermediate bound $\Pi \leq (2n)^{\pi(\sqrt{2n})/2}$. The least solutions are $n = 208$ ($k = 2$), $n = 3475$ ($k = 3$), $n = 8174$ ($k = 4$), against bounds 3.82, 10.22, 14.19.

Corollary 8 (what remains). *For fixed $k \geq 2$, #727 follows from the conjunction of*

(S) *a positive density of n for which $(n+1) \cdots (n+k)$ is $\sqrt{2n}$ -smooth, and*

(C) *a positive conditional probability, given (S), that $\kappa_p(n) \geq 2v_p$ holds for every $p \leq \sqrt{2n}$.*

Theorem 4 shows (S) is necessary, so no route can avoid it.

Condition (C) is of the same type as the carry conditions in Erdős #377, but is required only with positive probability rather than uniformly in n , which is a substantially weaker demand.

4 Erdős #374

For $a \geq 1$ let $w(a) \in \mathbb{F}_2^{(\text{primes})}$ have p -coordinate $v_p(a!) \bmod 2$. Since $v_p(a!) = v_p((a-1)!) + v_p(a)$ we have the ledger recursion $w(a) = w(a-1) + u(a)$, where $u(a)$ has p -coordinate $v_p(a) \bmod 2$. A product $a_1! \cdots a_k!$ is a perfect square iff $\sum_i w(a_i) = 0$ in $\mathbb{F}_2$. Recall $F(m)$ is the least $k \geq 2$, if it exists, for which there are $a_1 < \cdots < a_k = m$ with $a_1! \cdots a_k!$ a square, and $D_k = \{m : F(m) = k\}$.

Theorem 9 (primes are excluded, with a reason). *If m is prime then $F(m)$ does not exist. Equivalently, no D_k contains a prime.*

Proof. Let $m = p$ be prime. For $a < p$ we have $v_p(a!) = 0$, so the p -coordinate of $w(a)$ is 0; whereas $v_p(p!) = 1$, so the p -coordinate of $w(p)$ is 1. Any admissible family has $a_k = p$ and $a_i < p$ for $i < k$, so the p -coordinate of $\sum_i w(a_i)$ equals 1 $\neq 0$, and the product cannot be a square. $\square$

Remark 10. *The mechanism is that a prime is the first index at which its own coordinate is switched on, so $w(p)$ is automatically independent of all its predecessors. In this language, $F(m)$ exists precisely when $w(m)$ lies in the $\mathbb{F}_2$ -span of $\{w(a) : a < m\}$, and $F(m) - 1$ is the least number of distinct predecessors summing to $w(m)$ — a shortest-zero-sum (minimum-weight) problem, not a counting one.*

Theorem 11 (the four-term criterion). *Let $2 \leq c < m - 1$. Then $(c-1)!c!(m-1)!m!$ is a perfect square if and only if cm is a perfect square, i.e. iff c and m have the same squarefree kernel.*

Proof. As above $w(t) + w(t-1) = u(t)$, so $w(c-1) + w(c) + w(m-1) + w(m) = u(c) + u(m)$. This vanishes in $\mathbb{F}_2$ iff $u(c) = u(m)$, which says exactly that c and m have the same set of primes to odd exponent, i.e. the same squarefree kernel, i.e. cm is a square. $\square$

Verified: 177 903 pairs (c, m) with $m < 600$ — the equivalence is exact.

Theorem 12 (non-squarefree m need at most four terms). *Let $m = si^2$ with s squarefree and $i \geq 2$, and suppose $m \geq 8$. Then $F(m) \leq 4$; explicitly $c = s(i-1)^2$ satisfies $2 \leq c < m-1$ and $cm = (si(i-1))^2$.*

Proof. $cm = s^2i^2(i-1)^2$ is a square, so Theorem 11 applies once $2 \leq c < m-1$. Since $i \geq 2$, $c = s(i-1)^2 \geq s \geq 1$, and $c \geq 2$ unless $s = 1, i = 2$, i.e. $m = 4 < 8$. Also $c/m = \left(\frac{i-1}{i}\right)^2 \leq \frac{4}{9}$, so $c \leq \frac{4}{9}m < m-1$ for $m \geq 2$. $\square$

Verified: the construction succeeds for all 1566 non-squarefree m with $8 \leq m \leq 4000$, with no ledger failure and no range failure.

Corollary 13 (a positive-density answer).

$$|(D_2 \cup D_3 \cup D_4) \cap \{1, \dots, n\}| \geq \left(1 - \frac{6}{\pi^2}\right)n + O(\sqrt{n}) = 0.3921\dots n + O(\sqrt{n}).$$

Since $D_2 = \{j^2 : j > 1\}$ has $O(\sqrt{n})$ elements, $|(D_3 \cup D_4) \cap \{1, \dots, n\}| \gg n$.

Proof. By Theorem 12 every non-squarefree $m \geq 8$ has $F(m) \leq 4$, hence lies in $D_2 \cup D_3 \cup D_4$. The non-squarefree integers up to n number $(1 - 6/\pi^2)n + O(\sqrt{n})$. $\square$

Verified: 1567 non-squarefree $m \leq 4000$, density 0.3917 against $1 - 6/\pi^2 = 0.3921$.

Theorem 14 (pronic m need only four terms). *Let $m = a(a+1)$ with $a \geq 3$. Then $(a-1)!(a+1)!(m-1)!m!$ is a perfect square, so $F(m) \leq 4$.*

Proof. By the pair identity, $w(a-1) + w(a) = u(a)$ and $w(a) + w(a+1) = u(a+1)$, so

$$w(a-1) + w(a+1) = u(a) + u(a+1) = u(a(a+1)) = u(m),$$

using multiplicativity of u . Also $w(m-1) + w(m) = u(m)$. Adding the two relations gives $w(a-1) + w(a+1) + w(m-1) + w(m) = 2u(m) = 0$ in $\mathbb{F}_2$. The four arguments $a-1 < a+1 < m-1 < m$ are distinct and positive for $a \geq 3$. $\square$

Note that this construction is *shorter* than the generic one of Theorem 16 precisely because the two consecutive pairs overlap: the argument a occurs in both $\{a-1, a\}$ and $\{a, a+1\}$ and cancels, removing two terms.

Proposition 15 (parity of a witness). *Let S be a witness for m , i.e. $\max S = m$ and $\sum_{a \in S} w(a) = 0$. Partition S into maximal runs of consecutive integers. A run of even length 2ℓ contributes $\sum_{i=1}^{\ell} u(\cdot)$, a sum of ℓ parity vectors of integers; a run of odd length contributes such a sum together with one residual $w(\cdot)$. Hence if $|S|$ is odd, at least one run has odd length, and the witness necessarily involves a value of w itself rather than only values of u .*

Proof. Group each run $\{t, t+1, \dots\}$ from the bottom in consecutive pairs; each pair $\{s-1, s\}$ contributes $w(s-1) + w(s) = u(s)$. A run of odd length leaves one element unpaired, contributing its own w . Parity of $|S|$ equals the parity of the number of odd-length runs. $\square$

This is the obstruction to shortening the six-term construction. All even-length shapes express the condition purely in terms of u , which is multiplicative and therefore controllable: Theorems 11, 12, 14 and 16 are all of this type. A five-term witness must, by Proposition 15, involve a bare $w(x)$, and since w is the cumulative sum $\sum_{t \leq x} u(t)$ rather than a multiplicative function, the requirement becomes a *collision* $w(x) = u(cm)$ rather than an identity. Collisions are generic but not constructible, and their failure is exactly what makes D_6 nonempty.

Measured: the shape $\{c-1, c, x, m-1, m\}$, i.e. $w(x) = u(cm)$, admits a solution for **all** 475 integers $m \leq 900$ with $\Omega(m) \geq 3$, and for 232 of the 259 semiprimes in the same range; the 27 semiprime failures contain every element of D_6 .

Measured (exact computation of F by meet-in-the-middle over the ledger, all composite $m \leq 1400$): the sets $D_2, \dots, D_6$ have sizes 29, 80, 380, 250, 14 up to $m = 900$ and continue in the same proportions. Every element of D_5 is squarefree, as Theorem 12 requires. Every element of D_6 up to 1400 is a *semiprime*:

527 = 1731, 611 = 1347, 713 = 2331, 731 = 1743, 779 = 1941, 893 = 1947, 923, 1003, 1037, 1271, 1273, 1343, 1347,

the least being 527, matching the value recorded by Erdős and Graham; and every $m \leq 1400$ with $\Omega(m) \geq 3$ satisfies $F(m) \leq 5$. If the containment $D_6 \subseteq \{pq\}$ holds in general then $|D_6 \cap \{1, \dots, n\}| \ll n \log \log n / \log n$ by Landau, hence $o(n)$, answering the growth question for D_6 in the negative; this containment is measured here, not proved.

Theorem 16 (an explicit six-term construction). *Let $m = ab$ with $2 \leq a < b$, $b \leq m-2$ and $b \geq a+2$. Then*

$$(a-1)! a! (b-1)! b! (m-1)! m! \quad \text{is a perfect square,}$$

and the six arguments $a-1 < a < b-1 < b < m-1 < m$ are distinct. Hence $F(m) \leq 6$.

Proof. For any $t \geq 1$ we have $t! (t-1)! = t \cdot ((t-1)!)^2$, so in $\mathbb{F}_2$ coordinates $w(t) + w(t-1) = u(t)$, the parity vector of t . Applying this to $t = a, b, m$,

$$w(a-1) + w(a) + w(b-1) + w(b) + w(m-1) + w(m) = u(a) + u(b) + u(m).$$

Since $m = ab$, the exponent of each prime p in m is $v_p(a) + v_p(b)$, so $u(m) = u(a) + u(b)$ in $\mathbb{F}_2$ and the right-hand side vanishes. By the criterion of §4 the corresponding product of factorials is a square. The hypotheses $2 \leq a$, $b \geq a+2$ and $b \leq m-2$ make the six arguments strictly increasing and positive. $\square$

Verified: the construction succeeds for 3430 of the 3448 composite $m \leq 4000$. The 18 exceptions are $m = 6$ and the prime squares 9, 25, 49, 121, 169, 289, 361, 529, $\dots$ — precisely the composites admitting no factorisation $m = ab$ with $b \geq a+2$ and $b \leq m-2$.

Corollary 17. *$F(m) \leq 6$ for every composite m that is neither 6 nor the square of a prime.*

This recovers, by explicit construction, the bound $D_k = \emptyset$ for $k > 6$ recorded by Erdős and Graham, and exhibits the witness: the six-term family $\{a-1, a, b-1, b, m-1, m\}$ attached to any sufficiently separated factorisation of m . The excluded cases are exactly those with

no such factorisation, and the prime squares $m = p^2$ are handled separately by $D_2 = \{n^2 : n > 1\}$.

Measured. Computing the span incrementally for $m \leq 4000$: $w(m)$ fails to lie in the span for exactly 550 values of m , and $\pi(4000) = 550$. So in this range $F(m)$ *exists if and only if m is composite*, the forward direction being Theorem 9 and the converse remaining a measurement. Consequently $\bigcup_{k \geq 2} D_k$ has density 1, and since $D_2 = \{n^2 : n > 1\}$ has density 0 and $D_k = \emptyset$ for $k > 6$ [3], the sets D_3, D_4, D_5, D_6 partition a density-one set. Erdős's question — whether $|D_6 \cap [1, n]| \gg n$ — asks which of the four takes a positive share.

Remark 18 (register). **Proved:** Theorems 1, 4, 9, 7 and Lemmas 3, 5, 6; all are elementary and self-contained apart from the prime number theorem used in Theorem 7. Theorem 7 gives the first lower bound on n in terms of k for #727, namely $n \gg k^2 \log^2 k$. Theorem 16 and Corollary 17 give an explicit six-term witness showing $F(m) \leq 6$ for every composite m other than 6 and the prime squares; Theorems 11, 12 and Corollary 13 answer the growth question in part, giving $|(D_3 \cup D_4) \cap \{1, \dots, n\}| \gg n$ with explicit constant $1 - 6/\pi^2$. Theorem 9 re-proves a fact recorded in [3]; no priority is claimed, only the mechanism. **Measured:** all densities, the smoothness census, the span computation, and the converse of Theorem 9. **Not proved:** #727 for any $k \geq 2$, and the growth rates in #374. **Falsifiers:** a solution of #727 with $p_{\max} > \sqrt{2n}$ (would contradict Theorem 4 and hence a computation error); the $k = 2$ density declining toward 0 beyond $n = 10^7$; an $m \leq 10^5$ that is composite with $w(m)$ outside the span.

References

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